

Problem 2.1 A steel bar 50mm wide, 12mm thick and 300 mm long is subjected to an axial pull of 96 kN. Find the changes in the length, width, thickness and the volume of the bar. Take $E = 2 \times 10^5 \text{ N/mm}^2$ and Poisson's ratio $= 0.32$.

Solution: Longitudinal stress $= \sigma = \frac{P}{A} = \frac{96000}{50 \times 12} = 160 \text{ N/mm}^2$

$\therefore$ Longitudinal strain $= e = \frac{\sigma}{E} = \frac{160}{2 \times 10^5} = 0.0008$

Lateral strain = Poisson's ratio $\times$ Longitudinal strain

$= 0.32 \times 0.0008 = 0.000256$

$\therefore$ Increase in length of the bar $= \Delta l = e_l \times l = 0.0008 \times 300 = 0.24 \text{ mm}$

Decrease in width $= \Delta b = \text{Lateral strain} \times \text{Original width}$
 $= 0.000256 \times 12 = 0.0031 \text{ mm}$

Volumetric strain $= e_v = e_l + e_b + e_d$
 $= 0.0008 - 2 \times 0.000256 = 0.000288 \text{ (+ve)}$

$\therefore$ Increase in volume $= e_v \times V$
 $= 0.000288 \times 300 \times 50 \times 12 \text{ mm}^3 = 51.84 \text{ mm}^3$

Problem 2.2 A bar uniform rectangular section A is subjected to an axial tensile load P.

Show that the volumetric strain is given by $e_v = \frac{P}{AE} \left(1 - \frac{2}{m} \right)$ where E is the Young's

Modulus and $\frac{1}{m}$ is the Poisson's ratio.

Solution: Strain of the length $= e_l = \frac{P}{AE}$

$\therefore$ Lateral strain $= \text{Strain of width} = \text{Strain of depth}$
 $= -\frac{1}{m} e_l = -\frac{P}{mAE}$

$\therefore$ Volumetric strain $= \text{Strain of length} + \text{Strain of width} + \text{Strain of depth}$

$\therefore e_v = \frac{P}{AE} - \frac{P}{mAE} - \frac{P}{mAE}$

$\therefore e_v = \frac{P}{AE} \left(1 - \frac{2}{m} \right)$

Problem 2.3 A steel rod 5 meters long and 20 mm diameter subjected to an axial tensile load of 45 kN. Find the change in length, diameter and the volume of the rod. Take E_{steel}

$= 2 \times 10^5 \text{ N/mm}^2$ and Poisson's ratio $= \frac{1}{4}$.

Solution : Area of the rod = $A = \frac{\pi}{4} \times 20^2 = 314.2 \text{ mm}^2$

$\therefore$ Tensile stress $\sigma = \frac{P}{A} = \frac{45000}{314.2} = 143.2 \text{ N/mm}^2$

$\therefore$ Strain of length = $\frac{\sigma}{E} = \frac{143.2}{2 \times 10^5} = 0.000716 \text{ (+ve)}$

$\therefore$ Increase in length = $0.000716 \times 5000 = 3.580 \text{ mm (+ve)}$

Lateral strain = Strain of diameter = $\frac{\sigma}{mE} = \frac{1}{4} \times 0.000716 = 0.000179 \text{ (-ve)}$

$\therefore$ Decrease in diameter = $0.000179 \times 20 = 0.000358 \text{ mm}$

Volumetric strain = $\frac{\sigma}{E} - \frac{2\sigma}{mE} = 0.000716 - 2 \times 0.000179 = 0.000358 \text{ (+ve)}$

$\therefore$ Increase in volume = $0.000358 \times \frac{\pi}{4} \times 20^2 \times 5000 = 562.06 \text{ mm}^3$

Problem 2.4 A material has a Young's Modulus of $1.30 \times 10^5 \text{ N/mm}^2$ and a Poisson's ratio of 0.25. Calculate the modulus of Rigidity and the Bulk Modulus.

Solution: $E = 1.30 \times 10^5 \text{ N/mm}^2$
 $\frac{1}{m} = 0.25$

C = Modulus of Rigidity

K = Bulk Modulus

We know, $E = 2C \left(1 + \frac{1}{m}\right) = 2C (1 + 0.25) = 2.5 C$

$\therefore C = \frac{1.30 \times 10^5}{2.5} = 0.52 \times 10^5 \text{ N/mm}^2$

We know, $E = 3K \left(1 - \frac{2}{m}\right) = 3K (1 - 2 \times 0.25) = 1.5K$

$\therefore K = \frac{1.30 \times 10^5}{1.5} = 0.867 \times 10^5 \text{ N/mm}^2$

Problem 2.5 In a tensile test, a test piece 30 mm in diameter , 250 mm gauge length stretched 0.106 mm under a pull of 65 kN. In a torsion test, the same rod twisted 0.017 radian over a length of 250 mm, when a torque of 450 Nm was applied. Find the three elastic moduli and the Poisson's ratio for the material of the test piece.

Solution: $d = 30 \text{ mm}$; $l = 250 \text{ mm}$; pull = 65 kN = 65000 N

$$\text{Tensile stress} = \sigma = \frac{\frac{\pi}{4}(30)^2}{65000} = 91.95 \text{ N/mm}^2$$

$$\text{Tensile strain} = e = \frac{0.106}{250} = 4.24 \times 10^{-4}$$

$$\text{Modulus of Elasticity} E = \frac{\sigma}{e} = \frac{91.95}{4.24 \times 10^{-4}} = 2.169 \times 10^5 \text{ N/mm}^2$$

$$\text{Modulus of Rigidity} C = \frac{E}{2(1 + \mu)} = \frac{2.169 \times 10^5}{2(1 + 0.303)} = 0.832 \times 10^5 \text{ N/mm}^2$$

$$\text{We know, } E = 2C \left(1 + \frac{1}{m} \right)$$

$$\frac{1}{1 + m} = \frac{E}{2C} = \frac{2.169 \times 10^5}{2 \times 0.832 \times 10^5} = 1.303$$

$$\therefore \text{Poisson's Ratio} = \frac{1}{m} = 0.303$$

$$\text{We know } E = 3K \left(1 - \frac{2}{m} \right)$$

$$K = \frac{E}{3 \left(1 - \frac{2}{m} \right)} = \frac{2.169 \times 10^5}{3(1 - 2 \times 0.303)} = 1.835 \times 10^5 \text{ N/mm}^2$$

Problem 2.6 A rectangular block 250mm × 100mm × 80mm. is subjected to axial loads as follows:

500 kN tensile in the direction of its length

900 kN tensile on the 250 mm × 80 mm faces

1000 kN compressive on the 250mm × 100 mm faces

Assuming Poisson's ratio as 0.25, find in terms of the modulus of elasticity E of the material the strains in the direction of each force.

If $E = 2 \times 10^5 \text{ N/mm}^2$, find the value of the modulus of rigidity and bulk modulus for the material of the block. Also calculate the change in the volume of the block.

Solution: Fig 2.7 shows the block with its 250 mm, 100mm and 80 mm dimension parallel to X, Y and Z axes.

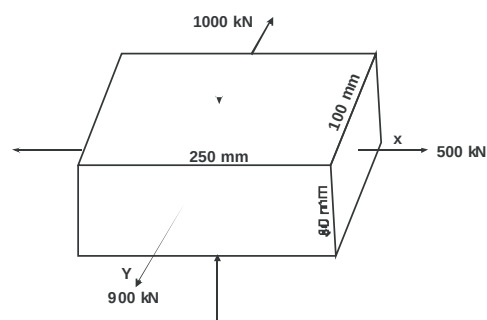


Fig. 2.7

The stresses in the direction of these axes are ,

$$\sigma_x = \frac{500 \times 1000}{100 \times 80} = 62.5 \text{ N/mm}^2 \text{ (tensile)}$$

$$\sigma_y = \frac{900 \times 1000}{250 \times 80} = 45 \text{ N/mm}^2 \text{ (tensile)}$$

$$\sigma_z = \frac{1000 \times 1000}{250 \times 100} = 40 \text{ N/mm}^2 \text{ (Compressive)}$$

The strains along the three principal directions are,

$$e_x = \frac{62.5}{E} - \frac{45}{mE} + \frac{40}{mE} = \frac{1}{E} \left(62.5 - \frac{45}{m} + \frac{40}{m} \right)$$

$$= \frac{1}{E} \left(62.5 - \frac{5}{m} \right) = \frac{1}{E} (62.5 - 5 \times 0.25) = + \frac{61.25}{E}$$

$$e_y = \frac{45}{E} - \frac{62.5}{mE} + \frac{40}{mE}$$

$$= \frac{1}{E} \left(45 - \frac{22.5}{m} \right) = \frac{1}{E} (45 - 22.5 \times 0.25) = + \frac{39.375}{E}$$

$$e_z = - \frac{40}{E} - \frac{62.5}{mE} - \frac{45}{mE} = - \frac{1}{E} \left(40 + \frac{107.5}{m} \right)$$

$$= - \frac{1}{E} (40 + 107.5 \times 0.25) = - \frac{66.875}{E}$$

$$\text{Volumetric strain} = e_v = e_x + e_y + e_z$$

$$= \frac{1}{E} (61.25 + 39.375 - 66.875) = + \frac{33.75}{E} = \frac{33.75}{2 \times 10^5}$$

$$\therefore \text{Increase in volume} = e_v \times V = \frac{33.75}{2 \times 10^5} \times (250 \times 100 \times 80) \text{ mm}^3 = 337.5 \text{ mm}^3$$

We know the following relation between the Young's Modulus E and the modulus of rigidity C and the bulk modulus K.

$$E = 2C \left(1 + \frac{1}{m} \right)$$

$$\text{And } E = 3K \left(1 - \frac{2}{m} \right)$$

$$\therefore 2 \times 10^5 = 2C(1+0.25) = 3K(1-2 \times 0.25)$$

∴

$$C = 0.8 \times 10^5 \text{ N/mm}^2$$

And

$$K = 1.33 \times 10^5 \text{ N/mm}^2$$

Problem 2.7 A square prism of wood 50mm × 50mm in cross-section and 300 mm long is subjected to a tensile stress of 40 N/mm² along its longitudinal axis and lateral compressive stresses of 20 N/mm² on one set of lateral face and a lateral tensile stress of 10 N/mm² on the other set of lateral faces. Find the changed dimensions of the prism. Take Poisson's Ratio = 0.4 and Modulus of Elasticity = 2 × 10⁴ N/mm².

Solution: Fig. 2.8 shows the prism subjected to the given stresses.

The strain along the length, breadth and depth are given by

$$e_x = \frac{40}{E} - \frac{10}{mE} + \frac{20}{mE}$$

$$= \frac{1}{E} (40 - 10 \times 0.4 + 20 \times 0.4)$$

$$= \frac{44}{E} = \frac{44}{2 \times 10^4} = 2.2 \times 10^{-3}$$

$$e_y = \frac{10}{E} - \frac{40}{mE} + \frac{20}{mE}$$

$$= \frac{1}{E} (10 - 40 \times 0.4 + 20 \times 0.4)$$

$$= \frac{2}{E} = \frac{2}{2 \times 10^4} = 1 \times 10^{-4}$$

$$e_z = -\frac{20}{E} - \frac{40}{mE} - \frac{10}{mE} = -\frac{1}{E} (20 + 40 \times 0.4 + 10 \times 0.4)$$

$$= -\frac{40}{E} = -\frac{40}{2 \times 10^4} = 2 \times 10^{-3}$$

∴

$$e_v = e_x + e_y + e_z = 2.2 \times 10^{-3} + 1 \times 10^{-4} - 2 \times 10^{-3}$$

$$= 0.3 \times 10^{-3}$$

$$\text{Increase in length} = 2.2 \times 10^{-3} \times 300 = 0.66 \text{ mm}$$

$$\text{Increase in width} = 1 \times 10^{-4} \times 50 = 0.0050 \text{ mm}$$

$$\text{Decrease in depth} = 2 \times 10^{-3} \times 50 = 0.1 \text{ mm}$$

$$\text{Increase in volume} = 0.3 \times 10^{-3} \times 50 \times 50 \times 300 = 225 \text{ mm}^3$$

Problem 2.8 A bar of copper is 40 mm × 40 mm in section and is 120 mm long. It is subjected to a tensile load of 200 kN along the longitudinal axis and tensile loads of 500 kN and 400 kN on the lateral faces.

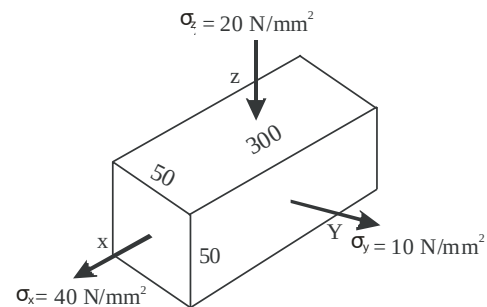


Fig. 2.8

- (a) Find the change in the dimensions of the bar and the change in volume.
 (b) Find also what axial longitudinal tensile load acting alone can produce the same longitudinal strain as in (a)

Take $E = 1 \times 10^5 \text{ N/mm}^2$ and $m = 0.3$

Solution: Fig. 2.9 shows the bar subjected to the given load system.

Let reference axes X, Y and Z be chosen as shown in the Fig. 2.9

Then stresses along these axes are

$$\sigma_x = \frac{200 \times 1000}{40 \times 40} = 125 \text{ N/mm}^2 \text{ (tensile)}$$

$$\sigma_y = \frac{500 \times 1000}{40 \times 120} = 104.17 \text{ N/mm}^2 \text{ (tensile)}$$

$$\sigma_z = \frac{400 \times 1000}{40 \times 120} = 83.33 \text{ N/mm}^2 \text{ (tensile)}$$

Strain along the X-axis

$$= e_x = \frac{\sigma_x}{E} - \frac{\sigma_y}{mE} - \frac{\sigma_z}{mE}$$

$$= \frac{1}{1 \times 10^5} (125 - 0.3 \times 104.17 - 0.3 \times 83.33)$$

$$= 0.0006875$$

$$\therefore \text{Increase in length} = 0.0006875 \times 120 \text{ mm}$$

$$= 0.0825 \text{ mm}$$

Strain along the Y-axis

$$= e_y = \frac{\sigma_y}{E} - \frac{\sigma_x}{mE} - \frac{\sigma_z}{mE}$$

$$= \frac{1}{1 \times 10^5} \{104.17 - 0.3(125 + 83.3)\} = 0.0004166$$

$$\therefore \text{Increase in width}$$

$$= 0.0004166 \times 40 \text{ mm} = 0.0167 \text{ mm}$$

$$= e_z = \frac{\sigma_z}{E} - \frac{\sigma_x}{mE} - \frac{\sigma_y}{mE} = \frac{1}{E} \left(\sigma_z - \frac{\sigma_x + \sigma_y}{m} \right)$$

Strain along the Z-axis

$$= \frac{1}{1 \times 10^5} \{83.33 - 0.3(125 + 104.17)\} = 0.0001458$$

$$\therefore \text{Increase in depth}$$

$$= 0.0001458 \times 40 \text{ mm} = 0.005832 \text{ mm}$$

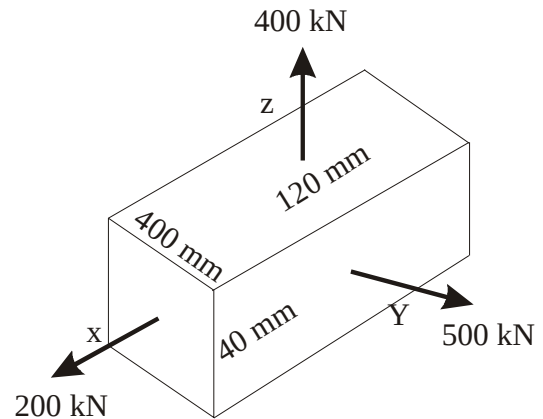


Fig. 2.9

Let P be the axial stress acting alone longitudinally i.e., along the X-axis to produce the same longitudinal strain .

$$\therefore \frac{P}{E} = 0.0006875$$

$$\therefore P = 0.0006875 \times 1 \times 10^5 = 68.75 \text{ N/mm}^2$$

$$\therefore \text{Longitudinal load} = 68.75 \times 40 \times 40 = 11000 \text{ N} = 110 \text{ kN}$$

Exercise

2.1 Define and explain lateral strain, longitudinal strain and Poisson's ratio.

2.2 Explain the 'state of simple shear'.

2.3 A bar of steel has rectangular cross section 30 x 20mm. Find the percentage decrease of area of cross section, when it is subjected to a tensile force of 120 kN in the direction of its length. Take $E = 2 \times 10^5 \text{ N/mm}^2$ and $\mu = 0.3$.

(Ans. 0.06%)

2.4 A bar of steel has rectangular cross section 30 x 30mm and 300 mm long. Find the change in volume, when it is subjected to a tensile force of 150 kN in the direction of its length. Take $E = 2 \times 10^5 \text{ N/mm}^2$ and $\mu = 0.3$.

(Ans. 90 mm³)

2.5 A bar of steel has rectangular cross section 40 x 40mm and 200 mm long. When it is subjected to a tensile force of 320 kN, 0.2 mm extension and 0.012 change in thickness were observed. Calculate the Poisson's ratio and the value of Young's modulus.

(Ans. 0.3, $2 \times 10^5 \text{ N/mm}^2$)

2.6 A steel cube of 50 mm side is subjected to a force of 10 kN (tension), 12.5 kN (compression) and 7.5 kN (tension) along x, y and z directions respectively. Determine the change in the volume of the block. Take $E = 2 \times 10^5 \text{ N/mm}^2$ and $\mu = 0.3$.

(Ans. 0.5 mm³)

2.7 Calculate the modulus of rigidity and bulk modulus of a cylindrical bar of diameter 25 mm and of length 1.2 m if the longitudinal strain in a bar during a tensile test is four times the lateral strain. Find the change in the volume when the bar is subjected to pressure of 120 N/mm^2 . Take $E = 1.2 \times 10^5 \text{ N/mm}^2$.

(Ans. $\mu=0.25$, $C=0.48 \times 10^5 \text{ N/mm}^2$, $K=0.8 \times 10^5 \text{ N/mm}^2$, change in volume=884 mm³)

2.8 For a material Young's modulus is given as $1.4 \times 10^5 \text{ N/mm}^2$ and Poisson's ratio 0.28. Calculate the Bulk modulus.

(Ans. $1.06 \times 10^5 \text{ N/mm}^2$)

2.9 For a material Young's modulus is given as $1.2 \times 10^5 \text{ N/mm}^2$ and modulus of rigidity is $4.5 \times 10^4 \text{ N/mm}^2$. Calculate the Bulk modulus and Poisson's ratio.

(Ans. $1.2 \times 10^5 \text{ N/mm}^2$, 0.33)

2.10 A steel bar of 320mm long, 40 mm wide and 30 mm thick is subjected to a pull of 250 kN in the direction of its length. Determine the change in volume. Take $E = 1.2 \times 10^5 \text{ N/mm}^2$ and Poisson's ratio 0.25

(Ans. 200 mm²)

2.11 A bar of 25 mm diameter is subjected to a pull of 40 kN. The measured extension on length of 200 mm is 0.085 mm and the change in diameter is 0.003 mm. Calculate the Poisson's ratio and the value of Young's modulus.

(Ans. 0.282, $1.917 \times 10^5 \text{ N/mm}^2$)